

NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

PROBLEM SET 5

Exchange Rate Regimes

1. An overvalued real exchange rate is a rate such that domestic goods are too expensive relative to foreign goods, net exports are too small, and by implication the demand for domestic goods is too low. This leads to difficult policy choices for the government and central bank. The equations that describe the economy are:

The IS curve:

$$Y = Y \left(\frac{\bar{E}P}{P^*}, G, T, i^* - \pi^e, Y^* \right) \quad (1)$$

The Phillips curves for the domestic and the foreign economy:

Domestic PC:

$$\pi - \bar{\pi} = (\alpha/L)(Y - Y_n) \quad (2)$$

Foreign PC:

$$\pi^* - \bar{\pi}^* = (\alpha^*/L^*)(Y^* - Y_n^*) \quad (3)$$

We make two critical assumptions. These are explore in parts (a) and (b). Then we move the analysis of the policy options when a country is experiencing an overvalued exchange rate.

- (a) We assume that the foreign economy is always in medium-run equilibrium. What are the implications of that assumption for foreign output and foreign inflation?
- (b) We assume that the domestic and foreign economies share the same anchored value for the level of expected inflation denoted so that $\bar{\pi} = \bar{\pi}^*$. What is the implication of that assumption once both the domestic and foreign economies are both in medium-run equilibrium?
- (c) Draw the IS-LM-UIP diagram for the case where the domestic country has an overvalued nominal exchange rate. What is the key feature of that diagram? Under fixed exchange rates without a devaluation, how does the economy return to its medium-run equilibrium?
- (d) Draw the IS-LM-UIP diagram for the case where the domestic country has an overvalued nominal exchange rate. Show how the economy can return to medium-run equilibrium when a devaluation is a policy choice.

- (e) Recall that the assumption has been made that interest rate parity holds, so $i = i^*$ at all times. Compared the returns on the domestic bond and returns on the foreign bond in the period of the devaluation. Will bond holders continue to believe there is a completely fixed nominal exchange rate? If bond holders believe that another devaluation is possible, what are the consequences for domestic interest rates?

2. An exchange rate crisis occurs when the peg (fixed exchange rate) loses its credibility. Bond holders no longer believe that next period's exchange rate will be this period's exchange rate.

Period	i_t	i_t^*	E_t	E_{t+1}^e
1		3	0.5	0.5
2		3	0.5	0.45
3		3	0.5	0.45
4		3	0.5	0.5
5	15%	3	0.5	0.4
6		3	0.4	0.4

- (a) Solve the uncovered interest parity condition for the value of the domestic interest rate in period 1.
- (b) In period 2, the crisis begins. Solve the uncovered interest parity condition for the value of the domestic interest rate in period 2.
- (c) The crisis continues in period 3. However, in period 4, the central bank and government resolve the crisis. How does this occur?
- (d) Unfortunately, in period 5, the crisis returns bigger and deeper than ever. Has the central bank raised interest rates enough to maintain uncovered interest rate parity? What are the consequences for the level of foreign exchange reserves?
- (e) How is the crisis resolved in period 6? Does this have implications for the future credibility of the central bank and the government?